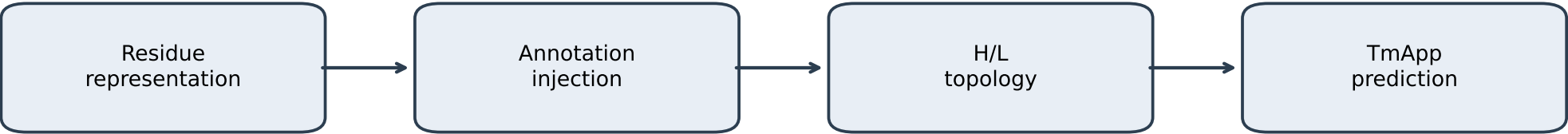


A. Overall factorial pipeline



Controlled comparison after frozen residue embeddings (or learned AA for Scratch).
REG = a learned chain-level token that gathers information from residues in the downstream Transformer.

B. Downstream H/L topologies (presentation IDs)

